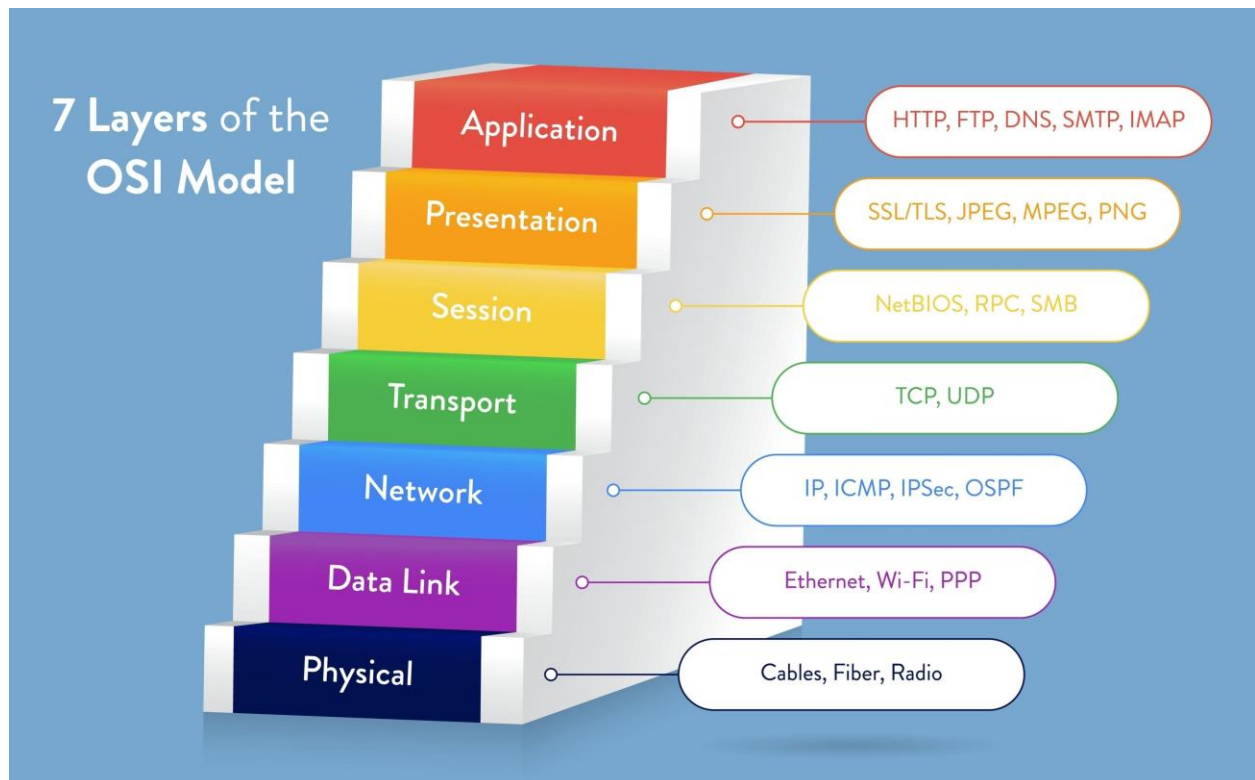


Question: 1

Draw an OSI Model diagram showing all 7 layers, and for each layer, write one real-world example (such as HTTP for Application, TCP for Transport, Ethernet for Data Link, cables for Physical, etc.)?

Answer:



1. OSI Model Diagram (Top to Bottom)

-
- **[Layer 7] Application Layer**

→right arrow

→

Example: HTTP / HTTPS (Used by web browsers like Chrome to load websites).

- **[Layer 6] Presentation Layer**

→right arrow

→

Example: SSL / TLS (Handles encryption and data formatting so it is readable).

- **[Layer 5] Session Layer**

→right arrow

→

Example: NetBIOS (Manages, opens, and terminates communication connections between devices).

- **[Layer 4] Transport Layer**

→right arrow

→

Example: TCP (Ensures reliable, error-checked delivery of data packets).

- **[Layer 3] Network Layer**

→right arrow

→

Example: IP Address / Routers (Determines the physical path data takes to reach its destination).

- **[Layer 2] Data Link Layer**

→right arrow

→

Example: Ethernet / MAC Address (Handles data transfer between two directly connected nodes on the same local network).

- **[Layer 1] Physical Layer**

→right arrow

→

Example: Cables & Hubs (The physical hardware, wires, and electrical pulses moving raw bits).

-

To test your understanding, try a quick practice problem:

Which layer of the OSI model is responsible for **routing data packets** across different networks using **IP addresses**?

a) Layer 2: Data Link Layer

- b) Layer 3: Network Layer
- c) Layer 4: Transport Layer
- d) None of the above

Question: 2

List three apps you use daily (for example: Instagram, Zomato, or WhatsApp) and for each, pick one network activity (like sending a message or placing an order) and explain which OSI layers are involved in that activity

1. WhatsApp – Activity: Sending a Text Message

- **Application Layer (Layer 7):** WhatsApp opens the chat UI and processes your text input when you hit "Send."
- **Presentation Layer (Layer 6):** Secures your text by applying end-to-end encryption so only the receiver can decrypt it.
- **Transport Layer (Layer 4):** Uses the **TCP protocol** to break the encrypted message into packets, guaranteeing they arrive intact and in the right sequence.

2. Instagram – Activity: Loading a Photo Reel/Feed

- **Application Layer (Layer 7):** Uses the **HTTPS protocol** to fetch and download the image/video data from Instagram's media servers.
- **Presentation Layer (Layer 6):** Decompresses raw image or video data formats (like JPEG or MP4) so they display correctly on your phone screen.
- **Network Layer (Layer 3):** Routes the media packet requests from your phone over the internet using **IP addresses**.

3. Zomato – Activity: Tracking Real-Time Delivery Location

- **Application Layer (Layer 7):** The app interface fetches live *GPS* coordinate streams using *API* web calls.

- **Transport Layer (Layer 4):** Often uses the **UDP protocol** for fast, real-time location streaming where speed is favored over resending dropped packets.
- **Data Link & Physical Layers (Layers 2 & 1):** Packages data into radio frames and transmits them wirelessly via your phone's **Wi-Fi or 4G/5G mobile hardware**.

To test your understanding, try a quick practice problem:

When an app like WhatsApp encrypts your data to keep it secure, which OSI layer is handling that specific **data encryption and formatting**?

- a) **Layer 7: Application Layer**
- b) **Layer 6: Presentation Layer**
- c) **Layer 5: Session Layer**
- d) **None of the above**

Question: 3

Install and open Wireshark on your computer, start a capture, then visit any website (for example, www.example.com) in your browser. Stop the capture and save the PCAP file?

Answer: **1. Step-by-Step Execution Guide**

- **Download & Install:** Go to the official Wireshark website to download and install the software for Windows or Mac.
- **Select Network Interface:** Open Wireshark. Double-click your active network connection from the startup list (usually labeled **Wi-Fi** or **Ethernet** with a jagged live traffic line).
- **Start Capture:** The packet capture begins automatically once you click the interface. You will see colored packets filling up the screen.
- **Generate Traffic:** Open your web browser (Chrome, Edge, or Firefox) and visit a simple site like www.example.com or any preferred web address.
- **Stop Capture:** Return to Wireshark and click the red square **"Stop" button** in the top-left corner toolbar.

- **Save PCAP:** Click **File** → **Save As**, name your file, select the default format (**Wireshark/tcpdump pcapng** or **.pcap**), and hit save.

2. What to Write in Your Assignment Submission

- **Network Interface Used:** State whether you used a **Wi-Fi** or **Ethernet** interface to capture the logs.
- **Target Website:** Note down the exact website domain name and its resolved destination IP address seen in the logs.
- **Primary Protocols Captured:** Mention that you observed **DNS** (for domain resolution), **TCP** (handshake), and **TLS/HTTP** (website data transport) in your capture list.

Question: 4

In Wireshark, find a captured HTTP request packet from your browsing session, take a screenshot of the packet details, and annotate which parts of the packet correspond to the Application, Transport, Network, and Data Link layers.

<elm>Hint: Use the packet details pane in Wireshark to expand each protocol section (HTTP, TCP, IP, Ethernet). </elm>?

Answer:

1. How to Find the Packet

- **Filter:** Type **http** in the top filter bar and press Enter.
- **Selection:** Click on any packet that says **GET** or **POST** in the Info column.

2. Packet Details Layer Breakdown (For Annotation)

Open the **Packet Details Pane** (middle window) and expand the sections to find these exact mappings:

- **Data Link Layer (Layer 2):** Annotated at the **Ethernet II** section. It contains the Source and Destination **MAC Addresses**.
- **Network Layer (Layer 3):** Annotated at the **Internet Protocol Version 4 (IPv4)** section. It contains the Source and Destination **IP Addresses**.

- **Transport Layer (Layer 4):** Annotated at the **Transmission Control Protocol (TCP)** section. It contains the Source and Destination **Port Numbers** (like Port 80 or 443).
- **Application Layer (Layer 7):** Annotated at the **Hypertext Transfer Protocol (HTTP)** section. It contains the website host details, user-agent, and the requested file path.